

Yanxin Luo

Tel: 984-317-4126 | luoyx28@hotmail.com | [LinkedIn](#) | [GitHub](#)

EDUCATION

Monroe Community College Rochester, NY

Computer Science

August 2026 - Present

- Relevant Courses: Calculus II, University Physics I, Programming Embedded Microcontrollers in C and Assembly

Durham Technical Community College

Durham, NC

Information Technology (Computer Science Focus)

August 2025 - July 2026

- GPA: 3.85/4.0
- Relevant Courses: JAVA, C++, Python, Calculus I, Web, Programming, and Database Foundations

Hengyang Normal University

Hengyang, China

BEng in Electronic Information Engineering

September 2013 - June 2017

- Relevant Courses: Digital Circuit Design, Analog Circuit Principles, Signal Analysis and Processing, Linux & C Programming, Data Structures and Algorithms in C

RESEARCH & PROJECTS EXPERIENCE

Wireless and Mobile Sensing, North Carolina State University

Raleigh, NC

Temporal Gait Event Timing from Low-Frame-Rate Video

March 2026 - Present

- Led the end-to-end engineering development, from the proof-of-concept application to a pose-based gait-event prediction model and its full training, evaluation, and visualization pipeline.
- Reduced temporal mean absolute error (MAE) by approximately 67% compared with baseline, providing the core experimental validation for the work accepted at the HumanSys 2026 Workshop at ACM MobiCom.

AI-Enhanced Wearable Interface for Meetings, Independent Research

Durham, NC

[SmartRing](#)

December 2025 - Present

- Designed and built a multimodal *SmartRing* prototype on nRF52840, integrating IMU-based gesture control, BLE HID interaction, audio capture, embedded UI, and battery management.
- Developed a real-time AI pipeline using the Google Gemini APIs to transform live speech into meeting summaries, key decisions, and action items for hands-free documentation.

AI Hackathon, Duke University

Durham, NC

[MagicUp](#)

November 2024 - November 2024

- Built an Orange Pi 5-based smart mirror within a 48-hour hackathon, integrating camera, microphone, display, and a custom enclosure for real-time AR overlays.
- Developed an offline voice-activated AI workflow using Snowboy, face parsing, and OpenAI/Google APIs to generate personalized digital makeup effects.

Beginner Award of Generative AI Hackathon, Duke University

Durham, NC

[AI Language Learning Product "TalkwithMe"](#)

November 2023 - November 2023

- Drove the development of an AI-powered English learning app from concept to a functional prototype within 48 hours, using Azure Text-to-Speech and Pronunciation Assessment APIs.

Undergraduate Capstone Project, Hengyang Normal University

Hengyang, China

ARM11-Based WiFi Video-Controlled Robotic Vehicle

March 2017 - June 2017

- Designed and built an ARM11-based robotic vehicle with embedded Linux, real-time Wi-Fi video streaming, and mobile remote control.
- Improved real-time responsiveness on resource-constrained hardware through system-level optimization and driver adaptation, reducing video and control latency.

PUBLICATION

Fu, Y., **Luo, Y.**, Ndiaye, S., & Xu, C. (2026). *Out-of-clinic measurement of time-domain gait parameters from low-frame-rate videos*. Accepted to the HumanSys 2026 Workshop at ACM MobiCom.

WORK EXPERIENCE

Zenitek Intelligent Technology Co., Ltd.

Shanghai, China

Senior Embedded Software Engineer

December 2021 - August 2023

- Owned the embedded software development of an STM32F7-based power detection system within a 5-person team, rearchitecting 64-channel data acquisition, signal processing, and power computation to reduce the sampling cycle from 1s to 100 ms (10× faster).
- Optimized the 16-bit ADC measurement pipeline and server communication interface, improving power-detection accuracy by 3 percentage points while enabling stable real-time transmission of 64-channel measurements and device-status data.

Shanghai Quicktron Intelligent

Shanghai, China

Senior Embedded Software Engineer

April 2021 - December 2021

- Spearheaded the development, maintenance, and testing of radar calibration software for robotic panels, improving calibration accuracy by 38%.
- Developed warehouse robot control and communication modules with a 5-person motion-control team, improving response speed by 18% and resolving field failures across three provinces within one month.

Focused Photonics Inc.

Hangzhou, China

Embedded Software Engineer

October 2018 - April 2021

- Collaborated with a 10-person team to bring a dust-monitoring instrument from prototype to production, developing test and embedded software, resolving thermal and stability issues, and improving Modbus data transmission efficiency by 25%.
- Redesigned the multi-device communication architecture and user interface for an NO/SO₂ monitoring system, reducing manual verification effort by over 50% in factory deployments and operational costs by at least 20%.

Sunward Intelligent Equipment Co., Ltd.

Changsha, China

Embedded Software Engineer

July 2017 - September 2018

- Primarily responsible for hardware selection and software development of a beyond-visual-line-of-sight (BVLOS) drone communication module, reducing hardware costs by 15%.

- Redesigned multi-channel transmission control, reducing packet loss by approximately 25% and extending communication range by $5\times$ over the original module.

TEACHING & MENTORING EXPERIENCE

Learning Centers, Monroe Community College

Rochester, NY

Peer Tutor

September 2026 - Present

- Provide one-on-one and small-group tutoring in mathematics and programming, supporting students in developing problem-solving skills and independent learning strategies.

VOLUNTEER EXPERIENCE

NC Chinese Music Ensemble

Cary, NC

Software Engineer and Performing Musician (Flute/Guitar)

February 2026 - Present

- Independently developed and maintain the ensemble's official website and digital platform, supporting event, media content, and public engagement.
- Support nonprofit cultural outreach through both digital engagement and live performances, promoting Chinese traditional music across North Carolina.

University Electronics Club

Hengyang, China

Electronics Repair Technician

October 2013 - October 2016

- Applied hands-on electronics and circuit-level troubleshooting skills to diagnose and repair embedded and mobile devices to local communities and senior care centers.

SKILLS

Programming: C, C++, Python, Java, Linux C, Shell, HTML, JavaScript

Embedded Systems & MCUs: STM32, Freescale, ESP32, FreeRTOS, UCOSII

Software & Tools: ROS, Qt, Linux, Git, MATLAB

Languages: Mandarin (native); English (professional proficiency)